

OSCAR DJURIC

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PhD student at Imperial College London focusing on HPC for aeronautical simulations. Skilled in C++, Python, and developing data-driven engineering solutions.

EDUCATION

Imperial College London

London, UK

PhD, Department of Aeronautics, Computational Engineering

10/2026 - Present

- Project: Novel algorithms for load rebalancing of adaptive mesh simulations on heterogeneous clusters.
- Redistributing and rebalancing parallel simulation workloads.
- Industrial partners: McLaren F1, Rolls-Royce and UKAEA.
- Contributing towards the advanced development of the Nektar++ software.

University of Bristol

Bristol, UK

MEng, Mechanical Engineering

09/2022 - 06/2026

- 1st Class Honours.
- Valedictorian (Mechanical Engineering Performance Prize).
- Best Master's project (Bechtel Industrial Project Group Prize).
- IMechE Best Student Certificate & IMechE Best Project Certificate.
- Masters Project: Sensing, Imaging and Signal Processing.

Queensland University of Technology

Brisbane, Australia

Exchange Year

07/2024 - 07/2025

- GPA: **6.875 / 7**, achieving high distinction on 7 out of 8 units, and 1 distinction.

Harris Westminster Sixth Form

London, UK

A Levels

09/2020 - 07/2022

- Physics, Chemistry, Mathematics, Further Mathematics

PROFESSIONAL EXPERIENCE

Cloudflare

London, UK

SWE Intern

06/2026 - 09/2026

- Developed the Data Centre Security & Compliance Portal, a web-based tool which included new tools and automation workflows for the DCSC Team for completing their monthly and quarterly security checks.
- Implemented a custom parsing pipeline which handled over 95 different colocation client data formats, fully automated with no human interaction required.
- Designed a full Vision-Language Model pipeline to extract text data from messy PDFs and PNGs, fully running on Cloudflare hardware with asynchronous task completion.

Salute (previously Northshore)

London, UK

Data and Analytics Intern

07/2025 - 09/2025

- Developed and automated data pipelines using Python, Power BI, and Power Automate to support sustainability and engineering teams in data-driven decision-making for data centres.
- Designed and deployed algorithms for automated error detection, flagging, and correction of site data using normalisation, rule-based validation, and rolling averages. Fully removed the need for human data handling.
- Optimised ETL workflows and transitioned systems to an ELT architecture using SQL, increasing scalability and reliability while processing over 2 million data points per hour.

Queensland University of Technology

Brisbane, Australia

Concentrated Solar Power Research Intern

11/2024 - 04/2025

- Developed and validated ray-tracing and performance models for Linear Fresnel Reflectors (LFR) in concentrated solar power (CSP) plants.
- Incorporated environmental data on dust and weather to improve accuracy, achieving predictions within 1% of reference system performance for an industrial application.
- Implemented multi-threading to optimise computational efficiency, reducing simulation runtime by 79% compared to previous efforts.
- Delivered findings at SolarPACES 2025 (Spain) in the presentation "*Ray Tracing Enhanced Soiling Assessment of LFR Systems for Industrial Heat Applications*". Supporting materials available on [GitHub](#).

TECHNICAL SKILLS

- **C++:** HPC. **Python:** Prototyping, data analysis, machine learning. **SQL:** Databasing. **TypeScript:** UI/UX, web development.
- **Linux - ROCm/HIP - Kotlin - Bristol Hackathon - CI/CD - GitLab/GitHub**
- **Home Server:** Custom home server running proxmox to handle multiple VMs and LXC's. Self-hosting and personal compute cluster.
- **CAD, CFD & FEA.**
- **Advanced Telecommunications & RF Engineering**
- **PowerBI & Power Automate**
- **Bilingual in French**
- 7 years rowing at a national level, and now 4 years swimming at a regional level. I train 9-10 times a week while balancing my studies and full-time internships.